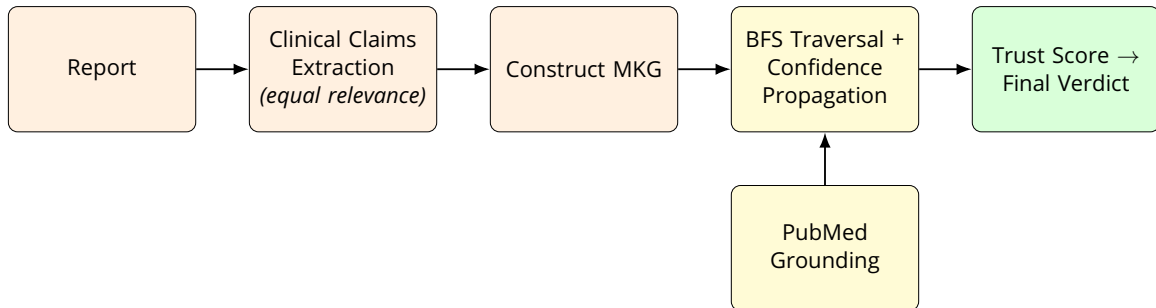


Why RadCheck RAG is needed?: LLM-generated radiology reports can hallucinate findings not actually in the image — sound fluent, wrong content. RadCheck RAG fixes this.

Core Idea: Hallucination detection needs both *external* evidence (is this finding medically documented?) and *internal* consistency checks (does this contradict another finding in the same report).

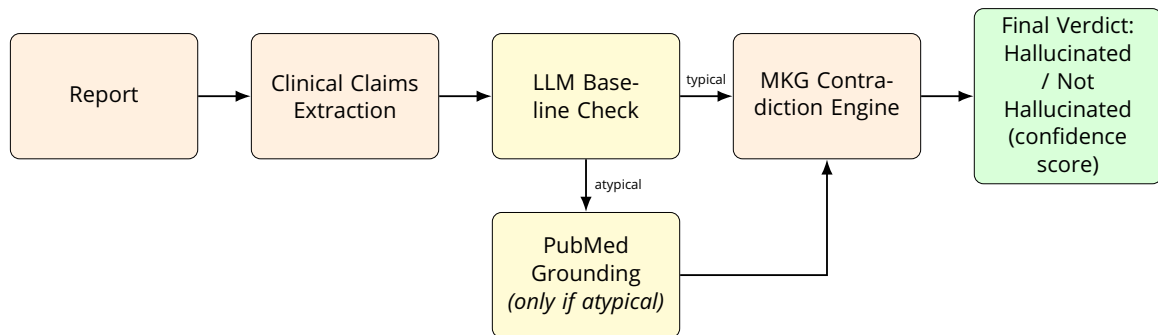
Previous Architecture



Drawbacks of Previous Architecture

- No explicit **contradiction detection** — only confidence propagation via BFS
- PubMed grounding applied **uniformly**, not selectively.
- No **bidirectional relationship modeling** between claims and KG nodes
- Lacks a structured **typical vs. atypical** decision path, increasing unnecessary computation

Architecture



Results

1. Clinical Claims Extraction

```
=====
Step 1: Report Ingestion & Extraction
=====

Input Report:
"The heart is of normal size with no abnormalities. The lungs are clear with no signs of consolidation or fluid accumulation. The structures around the lungs, including the mediastinum and hilar regions are normal. Overall, there are no acute or concerning findings."

Extracted 4 clinical claims:

Claim 1: "The heart is of normal size with no abnormalities"
Target: heart | Status: normal
Claim 2: "The lungs are clear with no signs of consolidation or fluid accumulation"
Target: lungs | Status: clear
Claim 3: "The structures around the lungs, including the mediastinum and hilar regions are normal"
Target: mediastinum and hilar regions | Status: normal
Claim 4: "no acute or concerning findings"
Target: overall | Status: normal
```

2. LLM Baseline Check

```
=====
Step 2: LLM Baseline Check (typical / atypical)
=====
```

```
Claim 1: [OK] TYPICAL
"The heart is of normal size with no abnormalities"
Reasoning: Standard description of a normal heart size and appearance.

Claim 2: [OK] TYPICAL
"The lungs are clear with no signs of consolidation or fluid accumulation"
Reasoning: Standard description of clear lungs without pathology.

Claim 3: [OK] TYPICAL
"The structures around the lungs, including the mediastinum and hilar regions are normal"
Reasoning: Standard observation of normal mediastinal and hilar anatomy.

Claim 4: [OK] TYPICAL
"no acute or concerning findings"
Reasoning: Common summary statement indicating a normal or non-urgent study.

Summary: 0 atypical / 4 total findings
```

3. PubMed Grounding & MKG Contradiction Engine

```
=====
Step 3: PubMed Grounding - only if atypical
=====

No atypical findings detected. Skipping PubMed grounding.

=====
Step 4: MKG Contradiction Engine
=====

Building Medical Knowledge Graph from report claims...
Nodes: 8 | Edges: 4
Heart --[has_finding]--> Claim_1
Lungs --[has_finding]--> Claim_2
Mediastinum And Hilar Regions --[has_finding]--> Claim_3
Overall --[has_finding]--> Claim_4

Running contradiction detection...

[OK] No internal contradictions detected.
```

4. Final Verdict

```
=====
Step 5: Final Verdict - Hallucinated / Not Hallucinated
=====
```

```
Claim 1: [PASS] NOT HALLUCINATED (Confidence: 1.00)
"The heart is of normal size with no abnormalities"
Baseline: typical | PubMed Grounded: false | Contradicted: false

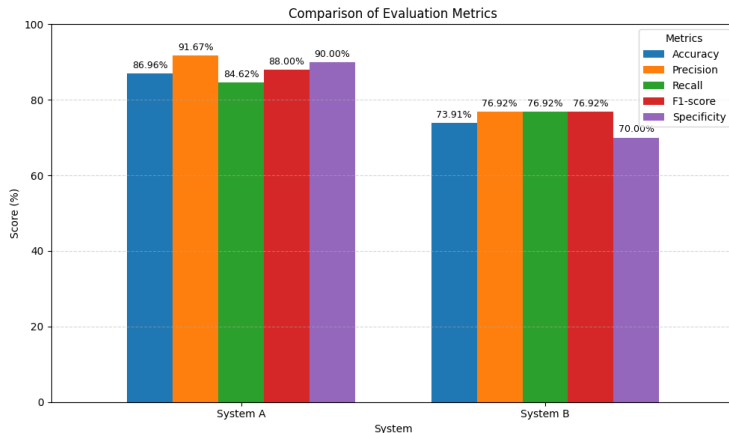
Claim 2: [PASS] NOT HALLUCINATED (Confidence: 1.00)
"The lungs are clear with no signs of consolidation or fluid accumulation"
Baseline: typical | PubMed Grounded: false | Contradicted: false

Claim 3: [PASS] NOT HALLUCINATED (Confidence: 1.00)
"The structures around the lungs, including the mediastinum and hilar regions are normal"
Baseline: typical | PubMed Grounded: false | Contradicted: false

Claim 4: [PASS] NOT HALLUCINATED (Confidence: 1.00)
"no acute or concerning findings"
Baseline: typical | PubMed Grounded: false | Contradicted: false

=====
OVERALL VERDICT: [PASS] REPORT IS CLINICALLY CONSISTENT
Average Confidence Score: 1.00
Graph Stats: 8 entities, 4 relations
=====
```

Architecture Comparison



Note: Evaluation was performed on a curated set of manually labeled clinical statements rather than raw dataset reports, since MIMIC-CXR/IU-Xray reports have no ground-truth contradiction/hallucination labels needed to compute precision, recall, and F1.